MAXIMILIAN LARTER

I am a plant ecophysiologist and an evolutionary biologist.
My primary research interest is understanding how plants respond and adapt to their environment, especially under climate change. I study functional traits and key physiological processes, and use statistics and models to predict the impacts of climate change on the performance, survival, and future distributions of wild and cultivated species.

RESEARCH EXPERIENCE

2025 — Postdoc – modelling urban heat islands and carbon storage
Sylvain Delzon's lab — INRAE, Bordeaux, France
- Model the impacts of management, development and land use changes on local temperature and carbon storage on the university campus.

2024 | 2021 — Postdoc – Plant hydraulics & trait trade-offs, forest ecology and biogeography
Sylvain Delzon's lab — INRAE, Bordeaux, France

2021 | 2019 — Postdoc – Herb hydraulics, positive root pressure and drought resistance in Brassicaceae
Frederic Lens's lab — Naturalis, Leiden, The Netherlands

2019 | 2017 — Postdoc – Evolution of the anthocyanin pathway in Iochrominae
Stacey Smith's lab — University of Colorado, Boulder, USA

2016 | 2012 — PhD – Evolution of cavitation resistance in conifers
Sylvain Delzon's lab — Université de Bordeaux, France
- *Supervisors*: Sylvain Delzon and Jean-Christophe Domec

EDUCATION

2025 | 2022 — Qualification CNU
Sections 66 Physiologie & 67 Biol. des populations et écologie

2016 | 2012 — PhD in evolutionary, functional and community ecology.
Thesis: Evolution of cavitation resistance in conifers
Université de Bordeaux — Bordeaux, France
- *Supervisors*: Sylvain Delzon and Jean-Christophe Domec

2011 — MSc – Terrestrial Ecosystem Functioning and Modelling
Université de Bordeaux — Bordeaux, France

2008 — BSc – Organismal Biology
Université d'Orléans — Orléans, France

SELECTED PUBLICATIONS

2025 — **M Larter**, G Charrier, S Delzon, W Hammond, A Baranger, C Bertrand, N Martin-StPaul, G Kunstler (2025) Weak global trade-off between frost and drought resistance in trees, *New Phytologist*. pdf

2024 — A Baranger, T Cordonnier, G Charrier, S Delzon, **M Larter**, N Martin-StPaul, G Kunstler (2024) Living on the edge - physiological tolerance to frost and drought explains range limits of 35 European tree species, *Ecography*. pdf

CONTACT

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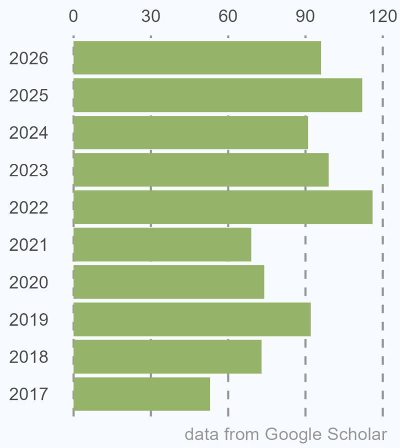
10 rue HG Wells,
33600 Pessac, France
August 5th, 1987. Derby (UK)

SKILLS

- Native speaker in **English** and **French**
- data analysis, stats, modelling, GIS (R, python, git, SAS)
- Plant hydraulics, physiology, gas exchange
- Molecular biology, phylogenetics

CITATIONS

935 citations, h-index 13



full publication list

Last updated on 2026-07-06.

2019 **M Larter**, A Dunbar-Wallis, AE Berardi, SD Smith (2019) Developmental control of convergent floral pigmentation across evolutionary timescales, *Developmental dynamics*, 248 (11), 1091-1100. pdf

2018 **M Larter**, A Dunbar-Wallis, AE Berardi, SD Smith (2018) Convergent evolution at the pathway level: predictable regulatory changes during flower color transitions, *Molecular biology and evolution*, 35 (9), 2159-2169. pdf

2017 **M Larter**, S Pfautsch, JC Domec, S Trueba, N Nagalingum, S Delzon (2017) Aridity drove the evolution of extreme embolism resistance and the radiation of conifer genus Callitris, *New Phytologist*., 215 (1), 97-112. pdf

2015 **M Larter**, TJ Brodribb, S Pfautsch, R Burlett, H Cochard, S Delzon (2015) Extreme aridity pushes trees to their physical limits, *Plant Physiology*, 168 (3), 804-807. pdf

2014 PS Bouche, **M Larter**, JC Domec, R Burlett, P Gasson, S Jansen, S Delzon (2014) A broad survey of hydraulic and mechanical safety in the xylem of conifers, *Journal of Experimental Botany*, 65 (15), 4419-4431. pdf

GRANTS AND FUNDING

- Active participation in grant writing
  ANR projects, Calls for Recherche Inter-disciplinaire et Exploratoire and Prématuration-Innovation at Université de Bordeaux, Fondation pour la Recherche et la Biodiversité IDEASHARE & DATASHARE

2020 Alberta Mennega Stichting fieldwork grant — Leiden (The Netherlands)
1,250€

2014 External mobility grant from the COTE Cluster of Excellence — Bordeaux (France)
3,000€

Research Exchange Program (Inbound) — Western Sydney University, NSW (Australia)
AU$3350

CONFERENCES AND PRESENTATIONS

2025 Poster - "Weak global trade-off between frost and drought resistance in trees" — Sevilla (Spain)
Xylem International Meeting XIM6

2022 Talk - "Trade off in cold and drought tolerance in trees" — Wurzburg (Germany)
Xylem International Meeting XIM5

2017 Talk - "Linking changes in gene expression to the macroevolution of flower color in Iochrominae" — Portland, Oregon
Evolution Meeting

2015 Talk - "The evolution of cavitation resistance in conifers and the case of world-record *Callitris*" — Bordeaux (France)
Xylem International Meeting XIM2

PROFESSIONAL SERVICE

- Guest Lectures (1h) — Bordeaux U. & Leiden U.
  MSc courses Plant Physiology, Methods in Biodiversity Analysis
- Supervision of several 6 month student projects — Bordeaux U. & Leiden U.
  1 BSc and 4 MSc - two leading to publications
- Regular reviewer for scientific journals
  New Phytologist, Tree Physiology, PCE, Evolution, American Journal of Botany, Plos One, BMC Evol. Biol.
- Postdoc and short term contract representative 2021-2022 — Bordeaux (France)
  Conseil d'Unité UMR Biogeco